V1 master diagram: how Ember works

Ember keeps what a project has decided and learned. It has five parts. The AI only suggests. Fixed checks make sure each suggestion is valid. You decide what is saved.

AI suggests | Automatic check (no AI) | You decide | Saved

WORDS USED ON THIS PAGE

Item
One thing Ember keeps: a Decision, Assumption, Finding, or Question.

Source
The original text you gave Ember, kept word for word.

Link
A confirmed connection between two items.

Status
Where an item stands: Active, Challenged, or Closed.

Ledger
The permanent list of every change ever made.

Knowledge base
All items, links, and the ledger, kept in one file on your computer.

HOW THE FIVE PARTS CONNECT

1 Capture
turns a source into items

2 Relate
links items to each other

3 Maintain
flags items that need a second look

4 Query
reads it all out

5 Safety
backs up, exports, and checks the file

All five parts work on one place: the knowledge base file on your computer. Parts 1, 2, and 3 save into it. Part 4 only reads. Part 5 copies it, writes it out as text, or checks it. The file also refuses invalid saves, as a second check after the automatic checks.

1 Capture

Turns a source into items. You end up with only the items you said yes to.

You decide 1

You bring a source

A chat, a message, or a document. Up to about 5 MB of text.

Automatic check 2

Ember stores it

A full copy is kept word for word. If you bring the same text again, Ember reuses the copy.

AI suggests 3

The AI suggests items

It reads the source. Each suggestion has a type, a short statement, and an exact quote from the source. It can also see up to 50 recent items. This step sends your text to an AI service on the internet.

Automatic check 4

Ember checks each suggestion

Is the type allowed? Is the statement filled in? Does the quote really appear in the source? Is it nearly a copy of an item you already have? That last check only gives a warning.

You decide 5

You confirm

For each suggestion: yes, no, edit, or quit. Nothing is saved until you say yes.

THE FOUR KINDS OF ITEM

Decision
A choice the team made. Example: use dark green buttons.

Assumption
Something we treat as true but have not proven. Example: users have fast phones.

Finding
Something we discovered. Example: blue buttons failed the contrast test.

Question
Something we do not know yet. Example: does dark green pass the test?

WHERE EVERY ITEM STANDS

Active
Still counts.

Challenged
Needs a second look.

Closed
Finished. A closed item always carries a reason, such as answered, found wrong, replaced by a newer item, or expired.

GOOD TO KNOW

- The AI can be wrong. That is why every suggestion waits for you.
- Ember has no command that deletes an item.
- You can store a source without using the AI at all.

Commands: `ember capture` (needs an AI key) and `ember capture-source` (no AI).

2 Relate

Connects a new item to the items it belongs with. You end up with only the links you said yes to.

Saved 1

A new item exists

It comes from Capture.

Automatic check 2

Ember picks items to compare

It ranks the other items by similar meaning, by items already connected, and by the same project label (or, with no label, a similar date). It skips pairs that already have every allowed kind of link.

AI suggests 3

The AI suggests links

It sees the new item and each candidate. It suggests which ones connect and what kind of link joins them, from the five kinds below. This step also sends text to an AI service.

Automatic check 4

Ember checks each suggestion

Nine checks run. For example: is this kind of link allowed between these two kinds of item? Is it a duplicate? Would it make a loop of replacements? Is the target already Closed?

You decide 5

You confirm

For each suggestion: yes, no, skip, or quit. A no is remembered, so that link is not suggested again. A skip leaves it open for later.

THE FIVE KINDS OF LINK

supports
A Finding backs up a Decision or an Assumption.

contradicts
A Finding goes against a Decision, an Assumption, or another Finding.

depends on
A Decision rests on an Assumption, a Finding, or another Decision. An Assumption rests on a Finding or another Assumption.

supersedes
A newer item replaces an older item of the same kind. The old one is closed.

answers
A Finding or a Decision answers a Question. The Question is closed.

GOOD TO KNOW

- The AI sees only the type and the statement of each item. It does not see your sources.
- Ember refuses any kind of link that is not allowed between two kinds of item, even if the AI suggests it.

Command: `ember relate --item N` or `ember relate --latest`. Needs an AI key and an optional extra package.

3 Maintain

Keeps items true when something changes. Only items directly connected to the change are marked.

Saved 1

Something changes

You confirm a link, or you close an item.

You decide 2

You run the check

You run the sweep. Closing an item by hand applies the rules at once.

Automatic check 3

Fixed rules run

Ember applies the four rules below. No AI.

Automatic check 4

Items change status

An item becomes Challenged or Closed. All changes are saved together or not at all. Each change adds a line to the ledger.

You decide 5

You review

For each Challenged item you decide: set it back to Active, close it, or leave it.

AI suggests (optional) 6

Old Questions

For a Question whose newest answer is at least 30 days old, the AI may suggest closing it. You confirm each one. At most 10 per run.

THE FOUR RULES

Rule 1: replaced

You confirm that a new item replaces an old one. The old item is Closed.

Rule 2: answered

You confirm that an item answers a Question. The Question is Closed.

Rule 3: conflict

You confirm that two items disagree. The item the link points at becomes Challenged.

Rule 4: knock-on

An item is closed as wrong or replaced. Items that depend on it become Challenged.

GOOD TO KNOW

- The effect spreads one step only. If B is Challenged because A was closed, an item that depends on B is not flagged. Check the Challenged list yourself.
- A status you set by hand is kept until newer links bring new evidence.
- By hand you can also undo a replacement, edit a reason or a source, and remove a link. A removed link is marked Rejected, never erased.

Commands: `ember maintain`, `close-item`, `set-status`, `reverse-supersedes`, `edit-item`, `remove-link`.

4 Query

Reads items back out. Read only. It changes nothing.

You decide 1

You pick a way to ask

The four ways are listed below.

Automatic check 2

Ember reads

It looks at the saved items and links. It never changes them. No AI.

Saved 3

An answer comes back

Structured text that people and other tools can read. Nothing is rewritten.

THE FOUR WAYS TO ASK

Current state
All items that still count. Challenged ones come first.

Topic search
Type a word. Items whose text matches come first. It matches words, not meanings.

Why
Follow the links from an item to see the reasons behind it.

History
See what changed: the old and new value, the date, and the cause.

Commands: `ember current-state`, `topic-search`, `why`, `history`.

5 Safety

Protects your knowledge. None of these commands changes your items.

Backup
Saves a restorable copy of the knowledge base file. It keeps the 3 newest older copies. To restore, copy the backup back.

Export
Writes everything as one plain text file you can read without Ember. The same data always gives the same file.

Health check
Reads the knowledge base file and reports whether it is healthy. It never changes anything.

GOOD TO KNOW

- Ember has no restore command on purpose. Restoring must never silently overwrite your live knowledge.
- An export cannot be restored into Ember. That is the backup's job.

Commands: `ember backup`, `ember export`, `ember db-check`.

Checked against the code on 2026-10-30. Every command is listed in `docs/commands.md`. The words are defined in `GLOSSARY.md`.